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TCP ports used by Pulse

This page provides a list of TCP ports opened by the components of Pulse, when running it locally with default setup (embedded Zookeeper, ActiveMQ and H2 database).

Pulse Server

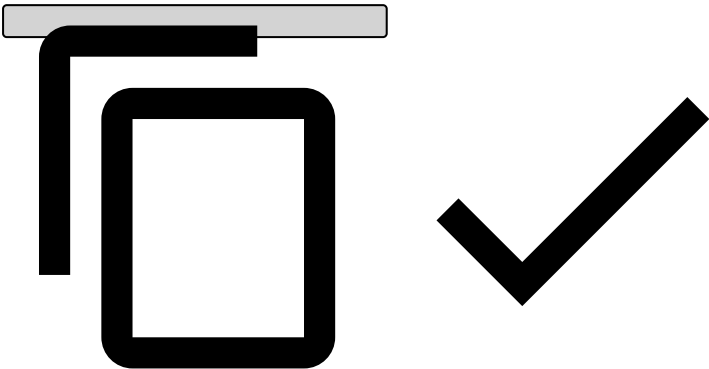
Fixed ports

- 2181: Zookeeper
- 61615: ActiveMQ
- 1099: Rmi Registry
- 1100: Remote Pulse Server object
- 9010: Jmx
- 8080: http
- 8443: https

Random ports

- H2 Database (Not used in production environment)
- ActiveMQ Jmx (Not used in production environment)
- JMX: (only when started with *startup-debug.sh* because *-Dcom.sun.management.jmxremote* is being used to launch the JVM)
- EhCache RMI port: can be configured on *conf/engine/shiro/ehcache.xml* by editing the *cacheManagerPeerListenerFactory* element and adding *properties="port=40001"*, for using port 40001.

```
<cacheManagerPeerListenerFactory>  
  class="net.sf.ehcache.distribution.RMICacheManagerPeerListenerFactory"  
  properties="port=40001"/>
```



App Engine

Fixed ports

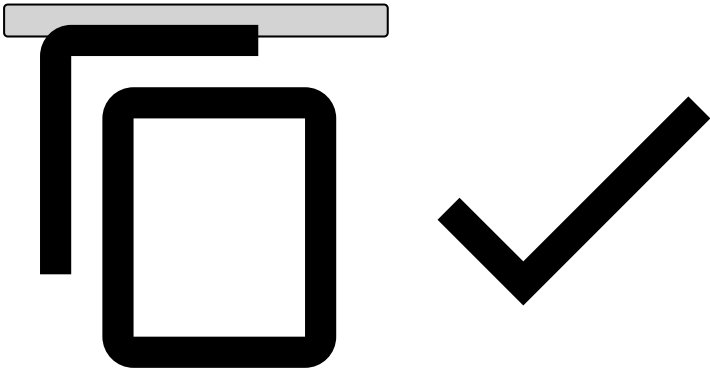
- 6990: App remote object and RMI registry

Random ports

- Jmx: configurable with:

```
# Global setting for all applications
pulse.global.appengine.jmxport=9011

# Configuration for a specific application
pulse.app.<app_slug>.appengine.jmxport=9013
```



- Other Jmx (because app engine is being started with `-Dcom.sun.management.jmxremote` a second random port is being open. This can probably be removed)
- On AppEngineProcess start, when getting the Pulse Server cluster leader remote reference, a RemoteExporter is being created and this opens a random RMI port as well. (This can potentially be fixed to use the configured RMI Object port. Not sure about the amount of effort because this code path is being used by other components)

Running processes on Spark (e.g. Data Source Analysis, Rules Snapshot Evaluation)

- A random port is opened (didn't have the chance to check where it comes from)

Conclusions

The table below summarizes the possible causes for TCP connections to random ports that are showing up on customer's logs and what could be a possible fix for each one:

Random port origin	Possible fix
EhCache RMI port	Configure as described above
App Engine Jmx	Configure as described above
App Engine Other Jmx	Remove <code>-Dcom.sun.management.jmxremote</code> from JVM args when launching application engine (if really not needed).

Typical customization

Typically, the following configurations are added/edited in `pulse.properties` to match the customer's environment.

```
pulse.manager.rmi.localregistryport=1099
pulse.global.services.modules.datascience.spark.forward.spark.driver.port=36560
pulse.global.services.modules.datascience.spark.forward.spark.blockManager.port=20800
pulse.global.services.modules.datascience.spark.forward.spark.port.maxRetries=20
pulse.manager.rmi.objectport=1100
```

